A pre-registered entanglement-forged H_4 benchmark on trapped-ion hardware: accurate noise modelling, incomplete error correction

Venkata Rao Allu[1, *]

[1] *Independent researcher*

(Dated: October 20, 2026)

We report a pre-registered benchmark of an entanglement-forged H_4 ground-state calculation on IonQ Forte Enterprise 1, a live trapped-ion processor. The analysis was frozen and publicly committed before any hardware data existed; the run was a single submission of 81 circuits at 1100 shots with device debiasing explicitly disabled. Two findings are in tension and we report both. A three-parameter analytic noise model, fixed in advance, predicted the *uncorrected* energy error to within 1% (23.84 against 24.06 kcal/mol) and the postselection retention to 0.4%. The correction built on that same model then failed: the oracle-free estimator returned 3.46 kcal/mol against 0.83 predicted, a factor of 4.16, missing chemical accuracy by 3.5× despite removing 85% of the raw error. Goodness of fit degraded from $\chi^2/\mathrm{dof} \approx 2$ on every noise model to 5.93 on hardware, in all 20 bootstrap resamples, so the failure was detectable from the hardware counts alone without reference to the known answer. Decomposing all 732 residuals by measurement group and Pauli weight localises the discrepancy to high-weight XY strings carrying the largest corrections and to a signed offset in one group. Refitting a single correction-magnitude parameter, blind to the reference energy, shows the model *under*-corrects by $\approx 1.8\times$; repairing this removes only a third of the excess and makes the energy slightly worse, so fit quality and accuracy are decoupled. We conclude that a noise model calibrated on raw observables must not be assumed predictive of corrected ones, that bootstrap resampling understates submission-to-submission variance, and that at 1100 shots the oracle-free estimator is shot-noise limited rather than noise limited, so mitigation alone cannot close the gap.

I. INTRODUCTION

Variational quantum eigensolvers [1, 2] are the most-studied near-term application of quantum processors, and the usual measure of success is whether a computed molecular energy falls within chemical accuracy, conventionally 1 kcal/mol (1.6×10^{-3} Hartree), of a classical reference. A large literature reports such agreement, most often on simulators with device noise models, sometimes on hardware with substantial error mitigation.

This paper takes a deliberately narrow and unglamorous question. Given a noise model that has been calibrated until it reproduces a device's *raw*, unmitigated behaviour, does the error correction built on that model transfer to the device? We answer it for one molecule, one device, one submission, with the analysis fixed in advance, and we report the answer whichever way it came out.

It came out negative in a specific and, we argue, informative way. The noise model's prediction of the raw error was nearly exact. Its prediction of the corrected error was wrong by more than a factor of four. The goodness-of-fit statistic flagged the failure — it tripled relative to every simulator rehearsal — which means the failure was detectable from the hardware data alone, without reference to the known answer. We regard that last point as the most portable result here: a mitigation pipeline should report a calibrated fit statistic, because that statistic distinguishes "the correction worked" from "the correction produced a number" without consulting an oracle.

Three practices made this possible and we recommend them. The analysis code was committed before the data existed, so no post hoc choice could be made. Every estimator was reported, including the one that failed. And the goodness-of-fit was computed and published alongside each energy.

II. THE ELECTRONIC-STRUCTURE ENGINE

All classical quantities were computed from a from-scratch implementation: contracted Gaussian basis evaluation, one- and two-electron integrals via the Obara–Saika recursions, restricted Hartree–Fock, integral transformation to the molecular-orbital basis, and the Jordan–Wigner mapping to a qubit Hamiltonian. Nothing is taken from a quantum-chemistry package at runtime. Correctness is established by comparison against PySCF [4] on the same geometries and basis, and the engine reproduces its Hartree–Fock and full configuration interaction (FCI) energies to numerical precision. We emphasise this only because it determines what the error bars mean: the reference energy against which all errors below are quoted is an exact FCI diagonalisation in the same basis, computed by the same code, not an external literature value.

The system is H_4 in a linear chain at 1.0 Å spacing, STO-3G, 4 electrons in 4 spatial orbitals, giving an 8-qubit Hamiltonian.

* alluvenkat11@gmail.com

III. METHODS

A. Entanglement forging

The 8-qubit problem is decomposed by entanglement forging [3], which writes the ground state as a Schmidt decomposition across a bipartition into two 4-qubit registers,

$$|\psi\rangle = \sum_{n=1}^{K} \lambda_n \, |u_n\rangle_\alpha \otimes |v_n\rangle_\beta, \tag{1}$$

so that the energy becomes a weighted sum of expectation values measured on one 4-qubit register at a time, over Schmidt basis states $|u_n\rangle$ and their pairwise superpositions $(|u_n\rangle + |u_m\rangle)/\sqrt{2}$. For H_4 at this geometry the Schmidt rank is effectively $K = 6$, giving 21 distinct preparation "slots" (6 singles and 15 pairs).

This is the first of several places where exact information enters the construction, and we flag each one explicitly. The state-preparation angles for each slot are obtained by classically fitting the circuit to reproduce the exact FCI Schmidt vectors, and the Schmidt coefficients λ_n used to assemble the energy are the exact ones. The calculation therefore measures whether hardware can reproduce a state whose identity is already known; it is not a blind variational search. Section VIII states the consequences.

B. Native-gate compilation

Circuits are compiled directly to IonQ's native gate set — GPi, GPi2, and the two-qubit ZZ interaction — rather than to a CNOT-based abstraction that the provider then retranslates. Angles are expressed in turns, with the ZZ angle restricted to $[0, 0.25]$. Where a negative interaction angle is required we use the identity $X{\cdot}\mathrm{ZZ}(\theta){\cdot}X = \mathrm{ZZ}(-\theta)$ with $X = \mathrm{GPi}(0)$, since the inverse method of the provider's gate objects does not negate the parameter.

C. General-commuting measurement grouping

The α-register Hamiltonian requires 273 Pauli expectation values. Grouping them into sets of mutually commuting — not merely qubit-wise commuting — operators and diagonalising each set with a Clifford circuit reduces this to 4 measurement groups. Across 21 slots this gives 84 circuits rather than 273, a 3.25× reduction. The grouping is obtained by exact graph colouring and verified to 6.7×10^{-16} against direct evaluation before use.

D. Ancilla-parity leakage detection

One ancilla qubit records the total parity of the 4-qubit register. The physical state lives in a fixed particle-number sector, so shots whose ancilla reads 1 have provably left that sector and are discarded. The retained fraction is reported for every run and is itself a prediction the noise model makes.

E. Analytic group-aware correction

The correction below is a forward analytic inverse of an assumed noise channel, in the spirit of zero-noise extrapolation and probabilistic error cancellation [5] and of the sparse Pauli–Lindblad models used to learn such channels on hardware [6], but with the channel parameters fixed from independent characterisation rather than learned on the target circuits.

For each slot and measurement group we compute, from the intended circuit and a three-parameter noise model, two quantities A_ℓ and B_ℓ per Pauli label ℓ: the expectation the noisy circuit would produce and the expectation the ideal circuit would produce. The measured value is then corrected multiplicatively,

$$\langle P_\ell \rangle_{\mathrm{corr}} = \langle P_\ell \rangle_{\mathrm{raw}} \cdot \frac{B_\ell}{A_\ell} \equiv \langle P_\ell \rangle_{\mathrm{raw}} \cdot r_\ell. \tag{2}$$

The three noise parameters — a ZZ depolarising rate $p_{ZZ} = 0.014593$, a GPi rate $p_{\mathrm{GPi}} = 0.000119$, and a GPi2 rate $p_{\mathrm{GPi2}} = 0.0004$ — were fixed before the hardware run from independent characterisation and never retuned on hardware data. The correction ratios r_ℓ for this experiment lie between 1.00 and about 1.20.

F. Three estimators

From the same counts we compute three energies, and we report all three.

Raw: a per-slot physical-state fit to the postselected but uncorrected expectation values.

Frame-free: the same per-slot fit applied to the corrected values of Eq. (2). No exact Schmidt vector enters the reconstruction. This is the oracle-free estimator and we treat it as the headline number.

Shared-frame: a single $K \times K$ orthogonal frame is fit jointly across all 21 slots, initialised at the exact frame, with the reconstruction basis spanned by the exact FCI Schmidt vectors. This estimator pools information across slots and is substantially tighter, but it depends on the exact state in two additional ways beyond the circuit construction. We report it because it was pre-registered, and we do not use it to claim chemical accuracy.

G. Goodness of fit

For the shared-frame fit we compute a shot-noise-weighted residual for every (slot, label) pair,

$$R_{s\ell} = \frac{\langle \hat{v}_s | P_\ell | \hat{v}_s \rangle - \langle P_\ell \rangle_{\mathrm{corr}}}{\sigma_{s\ell}}, \qquad \sigma_{s\ell} = |r_\ell| \sqrt{\frac{1 - \langle P_\ell \rangle^2_{\mathrm{raw}}}{N_{s\ell}}}, \tag{3}$$

with $N_{s\ell}$ the retained shot count, and report $\chi^2/\mathrm{dof} = \sum R^2/(n - N_\theta)$. Labels whose correction drives the value to near zero ($|r_\ell| < 0.01$) are excluded and counted. A well-specified model with correct error bars gives $\chi^2/\mathrm{dof} \approx 1$.

IV. PRE-REGISTRATION AND RUN CONDITIONS

The analysis script was committed before the hardware job was submitted and run unmodified on the returned counts. The conditions below were agreed with IonQ in advance and are stated because two of them materially affect the result.

- **Device debiasing disabled**, passed explicitly per circuit. With debiasing enabled the same job is priced at approximately $13,600 rather than $3,882. Debiasing also averages over qubit permutations, which would confound a correction model calibrated to specific gates.
- 1100 **shots per circuit**, fixed in advance.
- **A single submission**, with no resubmission permitted. No circuit was re-run and no outlier was discarded.
- 81 new circuits, plus 3 reused from an earlier job on the same device at 2000 shots, verified to rebuild with identical gate counts.

The job was submitted 2026-10-14 05:34 UTC and completed 2026-10-16 03:36 UTC. Cost was $3,882.21 by the provider's estimate endpoint and $3,875.18 by an independent rate-card calculation, a $7.03 discrepancy. Single-qubit gates accounted for 58.9% of the total (12,582 one-qubit and 1,302 two-qubit gates), which is worth noting because circuit-optimisation effort in this architecture is often directed at two-qubit count alone.

Two pre-registered gates were checked before analysis: the Schmidt sign convention reproduced stored ideal data to 3.2×10^{-5} (limit 10^{-3}), and all 84 required circuits were present.

TABLE I. Energy error against exact FCI, in kcal/mol, and shared-frame goodness of fit. Chemical accuracy is 1.0. Only the last row is a physical device.

	Raw	Frame-free	Shared-frame	χ^2/dof
ideal model	1.96	0.048	0.117	≈ 1
aria-1 model	22.53	1.041	0.707	≈ 2
forte-1 model	24.06	0.832	0.394	2.68
Forte Enterprise 1	**23.84**	**3.46**	**0.86**	**5.93**

V. RESULTS

A. Hardware against noise models

Table I gives the three estimators on hardware and on three noise models run through the identical pipeline. We stress that *aria-1* and *forte-1* denote device noise *models* offered for simulation, not hardware; the Aria-1 device has been retired and no submission to it is possible. Only the final row is hardware.

The raw agreement is the strongest quantitative result in this work. The hardware returned 23.84 kcal/mol where the forte-1 model predicted 24.06, a ratio of 0.99. Postselection retention was 0.894 against 0.898 predicted. A three-parameter model, fixed in advance, reproduced both the uncorrected error of an 8-qubit forged calculation and its leakage rate on a live device.

The corrected agreement fails. The frame-free estimator returned 3.46 kcal/mol where the same model predicted 0.832, a factor of 4.16. The correction is not useless — it removes 85% of the raw error, $23.84 \to 3.46$ — but it does not reach chemical accuracy, missing by 3.5×.

B. Uncertainty

Uncertainties come from a nonparametric bootstrap in which every circuit's counts are resampled at its own retained shot total and the entire pipeline, including the frame fit, is re-run ($N = 20$):

$$\begin{aligned} \text{raw} &: 19.49 \pm 5.65, \quad [14.48,\ 28.90]_{5-95} \\ \text{frame-free} &: 2.92 \pm 1.90, \quad [0.39,\ 6.47]_{5-95} \\ \text{shared-frame} &: 1.055 \pm 0.419, \quad [0.42,\ 1.87]_{5-95} \end{aligned}$$

The shared-frame point estimate of 0.864 lies below 1 kcal/mol, but its bootstrap distribution is centred *above* the threshold and 13 of 20 resamples exceed it. We therefore state that chemical accuracy is *consistent with* this dataset on the shared-frame estimator, and is not established by it. The frame-free estimator exceeds 1 kcal/mol in 16 of 20 resamples.

TABLE II. Mean squared residual $\langle R^2 \rangle$ by measurement group and Pauli weight. Uniform values near 1 would indicate a well-specified model.

	w=1	w=2	w=3	w=4
group 0	3.59	**10.14**	—	—
group 1	—	2.80	—	4.70
group 2	—	—	3.98	**13.58**
group 3	—	3.35	—	4.77

C. The fit statistic detected the failure

χ^2/dof was 0.9–1.2 on the ideal model, 1.6–2.7 on the two noisy models, and 5.93 on hardware, with every one of the 20 bootstrap resamples in $[6.41, 7.22]$. The excess is therefore a property of the data, not of one fit. Because this statistic requires no reference energy, the inadequacy of the correction model was visible from the hardware counts alone. Any pipeline reporting a mitigated energy can report this number, and we suggest that one which does not should be treated as unaudited.

VI. LOCALISING THE DISCREPANCY

A. Residual decomposition

We re-derived the identical 732-term residual vector and tagged each term with its slot, measurement group, Pauli label and Pauli weight. Table II gives mean R^2 by group and weight. (The design measures each Pauli label in exactly one group, so cells are structurally empty rather than missing.)

Two structures stand out.

The residual grows with the magnitude of the correction applied. Binning by correction ratio r_ℓ: labels with $r_\ell \in [1.00, 1.05)$ give $\langle R^2 \rangle = 3.69$, while labels with $r_\ell \in [1.15, 1.25)$ give 8.51 and carry 48% of the total χ^2 from 33% of the terms. The twelve largest individual residuals are all weight-4 XY-type strings ($YYXX$, $YXXY$, $XYYX$, $XXYY$, $YYYY$, XXXX), each with $r_\ell \approx 1.19$.

Group 0 carries a signed offset rather than scatter. Its mean *signed* residual is -1.32 at weight 1 and -1.73 at weight 2. Stochastic misestimation averages to zero; a systematic one does not.

B. Testing the magnitude hypothesis

The first structure suggests a simple repair: that Eq. (2) overshoots, so that

$$\langle P_\ell \rangle_{\mathrm{corr}}(\alpha) = \langle P_\ell \rangle_{\mathrm{raw}} \left[1 + \alpha\,(r_\ell - 1)\right] \tag{4}$$

with $\alpha < 1$ would reduce χ^2. Here $\alpha = 1$ recovers the pre-registered pipeline and $\alpha = 0$ applies no correction. Crucially, α is selected by minimising χ^2/dof alone; the reference energy is never an input, preserving the blindness of every correction-parameter choice in this work. Because fitting α on the same residuals whose χ^2 is then quoted would reduce it by construction, we also fit α on half the slots and evaluate χ^2 on the held-out half.

Two technical points must be settled before the scan is meaningful.

The exclusion rule is α-dependent. The pre-registered pipeline discards a label when $|\langle P \rangle_{\mathrm{corr}} / \langle P \rangle_{\mathrm{raw}}| < 0.01$. Under Eq. (4) that quantity is $1 + \alpha(r_\ell - 1)$, which for the 24 labels with $r_\ell \approx 0$ is small *only* when $\alpha \approx 1$. Recomputing the rule per α therefore excludes 24 labels at $\alpha = 1$ and none elsewhere, so χ^2 at different α would be evaluated over different residual sets, and the pre-registered value would enjoy an exclusion unavailable to its comparators. We freeze the mask at its $\alpha = 1$ membership, so all values below are computed on the same 732 terms.

The χ^2 objective has poles in α. The pre-registered pipeline propagates shot noise through the correction as $\sigma_{s\ell} \propto |1 + \alpha(r_\ell - 1)|$, which is correct for one given correction but vanishes at $\alpha = 1/(1 - r_\ell)$. In this dataset 26 labels have such a pole within $\alpha \in (0, 3]$: 24 at $\alpha = 1.0$ — the masked set above — and 2 at $\alpha = 1.2$, which accounts for an otherwise unexplained χ^2 spike at that point. The shot noise of a raw measurement does not change because the correction hypothesis changed, so we hold σ fixed at its $\alpha = 1$ value throughout the scan. With both controls the curve is smooth and monotone about its minimum.

The fit is converged. Holding the data fixed at $\alpha = 1$ and varying only the optimiser over 5 restart counts (4 to 64) and 5 random seeds, all 25 configurations returned χ^2/dof $= 5.929$ and 0.864 kcal/mol, with total spreads of 3.7×10^{-6} and 7×10^{-6} kcal/mol respectively. The pre-registered numbers are not an artifact of a particular restart draw. The same invariance shows the random restarts are inert: the fit is initialised at the exact frame and converges there, which is a further exact-state dependence.

With those controls in place, χ^2 falls as α *increases*:

	α	χ^2/dof	err (kcal/mol)
no correction	0.0	14.84	1.017
pre-registered	1.0	5.93	0.864
blind optimum	1.8	3.97	0.940
over-corrected	3.0	7.64	1.166

The hypothesis we set out to test — that the correction overshoots — is therefore refuted in its direction. The correction *under*-corrects by roughly a factor of 1.8. Held-out validation confirms this is not overfitting: fitting α on half the slots and scoring the other half improves χ^2/dof from 4.97 to 3.89 in one fold ($\alpha = 1.9$) and from 7.04 to 4.41 in the other ($\alpha = 1.7$).

Allowing a separate α_g per measurement group improves the in-sample fit further, to χ^2/dof $= 3.75$ at

$\alpha_g = (1.9, 1.4, 2.0, 1.3)$, and yields the lowest energy we obtain from this data, 0.637 kcal/mol. We do not advance that number. Under the earlier, α-dependent weighting the same model moved to $(2.0, 1.5, 2.1, 1.6)$ and 0.678 kcal/mol, and its held-out performance was no better than the global model's in one fold and worse in the other, with fitted vectors swinging between $(2.0, 1.7, 2.4, 1.9)$ and $(2.1, 1.3, 1.9, 1.1)$. Four parameters are not determined by this dataset, and the group structure is not stable enough to report as a result. The supported statement is the single global rescaling; the group structure is suggestive only. We note that the two groups demanding the largest correction, groups 0 and 2, are the two the residual decomposition independently flagged as worst.

Three conclusions follow, and the third is unwelcome. The correction model's magnitude is genuinely miscalibrated, in the opposite direction to our hypothesis. Repairing it removes only about a third of the χ^2 excess, $5.93 \rightarrow 3.97$ against a target near unity, so the misspecification extends beyond a scalar factor. And the better-fitting correction returns a *worse* energy, 0.940 against 0.864. Fit quality and accuracy are decoupled here, which bounds the claim of Sec. V C: χ^2 is a reliable detector of model misspecification, and not a proxy for accuracy.

A final observation constrains how much of this matters. At $\alpha = 0$, with no correction applied at all, the shared-frame estimator returns 1.017 kcal/mol against 0.864 with the full correction. The entire analytic correction stack is worth 0.15 kcal/mol on that estimator; the joint frame fit supplies the rest. Since the frame fit is the component carrying the exact-state dependence, the shared-frame result rests primarily on oracle information rather than on the physics-based correction. The frame-free estimator, where the correction does the work unaided, is the one that missed by 4.16×.

VII. DISCUSSION

A. Raw accuracy does not imply corrected accuracy

The central methodological claim of this paper is narrow and, we believe, well supported by the single comparison above: a noise model validated against raw device behaviour was wrong by 4.16× once its own correction was applied. The raw expectation value is a smooth, heavily averaged functional of the noise channel; the correction is a ratio of two such functionals, and ratios amplify misspecification. A noise model's agreement on uncorrected observables should therefore be treated as a necessary and distinctly insufficient condition for trusting its mitigation. Where a projection is needed, the corrected-observable prediction should be read as an optimistic bound.

B. Bootstrap understates submission variance

Before the hardware run, bootstrap resampling of earlier data predicted a frame-free maximum of 0.90 kcal/mol and a 100% pass rate at 1 kcal/mol for this design. Fresh, independent simulator submissions of the identical design at the same shot count returned up to 2.46 kcal/mol. Bootstrap resamples a single submission and therefore captures shot noise but not drift between submissions; the latter was the larger term. Reported bootstrap intervals in this setting should be regarded as lower bounds on the true spread.

C. A shot-noise floor, not a noise floor

The most consequential constraint we encountered is not device error. On one fresh 1100-shot submission the *zero-noise* ideal simulator itself returned 1.225 kcal/mol on the frame-free estimator. With no device error of any kind, the estimator missed chemical accuracy from counting statistics alone. Improved mitigation cannot repair this. Scaling as $N^{-1/2}$ from the worst observed value, pushing the frame-free estimator below 1 kcal/mol would require roughly 6.6×10^3 shots per circuit, which on the measured rate card costs about $23,400 for this design; below 0.5 kcal/mol, about 2.7×10^4 shots and $94,000. The productive direction is therefore a lower-variance estimator, not a larger shot budget.

D. Practical notes

Three observations may help others budgeting work on this architecture. Single-qubit gates were 58.9% of this job's cost, so optimising two-qubit count alone leaves most of the bill untouched. Device debiasing multiplies cost by roughly 3.5× and is on by default. Finally, we encountered three defects in the provider's Qiskit integration: two in multi-circuit result parsing and histogram scaling, since fixed upstream but not at the time in a tagged release, and one in which a cost-estimate field was read under the wrong key so that every estimate returned `None`. Pipelines that budget from that field will silently see no cost at all.

VIII. LIMITATIONS

We state these in the strongest form we can, because several of them bound the claims above severely.

This is not a blind variational calculation. The state-preparation angles are fit to the exact FCI Schmidt vectors, the Schmidt coefficients used to assemble the energy are exact, and the shared-frame reconstruction basis is spanned by the exact Schmidt vectors and initialised at the exact frame. What is measured is whether a noisy device plus mitigation can reproduce a *known* state. This

is a meaningful test of a mitigation stack and a weaker claim than locating an unknown ground state. This limitation was identified by an external reviewer reading the source repository, and the frame-free estimator — which removes the reconstruction-side dependence though not the circuit-construction one — is reported as the headline for that reason.

One molecule, one device, one submission. The single-submission condition that makes the pre-registration meaningful also prevents any statement about run-to-run hardware reproducibility. The bootstrap quantifies resampling uncertainty within one submission only.

Chemical accuracy is not demonstrated. The frame-free estimator missed by 3.5×. The shared-frame estimator's point estimate falls below the threshold but its bootstrap distribution is centred above it.

The analytic correction contributes little to the shared-frame estimator. Disabling it entirely changes that estimator from 0.864 to 1.017 kcal/mol. The joint frame fit, which is the component that depends on the exact Schmidt vectors and which is initialised at the exact frame, accounts for the remainder. The shared-frame number should therefore be read as a measurement of what oracle-assisted reconstruction can recover, not as a validation of the physics-based correction.

The three reused circuits were taken at 2000 shots from an earlier job on the same device, so the dataset is not uniform in shot count.

$K = 6$ **truncation.** The Schmidt rank is truncated; the resulting classical floor is far below the errors discussed here but is not zero.

IX. CONCLUSION

We pre-registered an analysis, ran one 81-circuit submission on a trapped-ion processor, and reported every estimator. A three-parameter noise model fixed in advance predicted the raw energy error to 1% and the post-selection retention to 0.4%. The correction built on that same model then missed the oracle-free target by a factor of 4.16, and the goodness-of-fit statistic rose from ≈ 2 to 5.93, flagging the failure without reference to the known answer. Residual decomposition localises the discrepancy to high-weight XY strings in the largest corrections and to a signed offset in one measurement group.

We did not reach chemical accuracy. We think the more useful output is the pair of negative results: that raw-observable calibration does not certify a correction model, and that at practical shot counts the oracle-free estimator is limited by counting statistics rather than by device noise. Both point away from more mitigation and towards lower-variance estimators and honest fit reporting.

ACKNOWLEDGMENTS

This work was supported by the IonQ Research Credits Program. We thank Vadim Karpusenko (IonQ) for technical guidance throughout, for identifying the native-gate compilation issue, for proposing the measurement-grouping reduction, and in particular for the review of this repository's source that identified the exact-state dependence described in Sec. VIII. The pre-registration conditions for the hardware run were set by IonQ and are reported verbatim.

In accordance with arXiv's policy on generative AI tools, we disclose that a large language model (Anthropic Claude) was used as a coding and drafting assistant during this work: for software implementation, for analysis scripting, and for preparing this manuscript. All scientific claims, experimental design decisions, pre-registration choices and interpretations are the author's own, and the author takes full responsibility for the content, including all numerical results, which were generated by the committed code and are reproducible from the public repository.

DATA AND CODE AVAILABILITY

All source code, the raw hardware counts for all 81 circuits, the frozen analysis script, and a complete iteration-by-iteration research log including every negative result are available at https://github.com/Venkatallu11/quantum-chemistry-vqe. The hardware result is reproduced by `task73_hw_analysis.py --counts task73_hw_counts.json --design full`, and the residual decomposition by `task82_hw_chi2_residual_breakdown.py`. Job identifier `01a138e7-c5fd-711e-bd12-41e418290d36`.

[1] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, and J. L. O'Brien, *A variational eigenvalue solver on a photonic quantum processor*, Nat. Commun. **5**, 4213 (2014).

[2] J. R. McClean, J. Romero, R. Babbush, and A. Aspuru-Guzik, *The theory of variational hybrid quantum-classical algorithms*, New J. Phys. **18**, 023023 (2016).

[3] A. Eddins, M. Motta, T. P. Gujarati, S. Bravyi, A. Mezzacapo, C. Hadfield, and S. Sheldon, *Doubling the size of quantum simulators by entanglement forging*, PRX Quantum **3**, 010309 (2022).

[4] Q. Sun *et al.*, *PySCF: the Python-based simulations of chemistry framework*, WIREs Comput. Mol. Sci. **8**, e1340 (2018).

[5] K. Temme, S. Bravyi, and J. M. Gambetta, *Error mitigation for short-depth quantum circuits*, Phys. Rev. Lett. **119**, 180509 (2017).

[6] E. van den Berg, Z. K. Minev, A. Kandala, and K. Temme, *Probabilistic error cancellation with sparse Pauli–Lindblad models on noisy quantum processors*, Nat. Phys. **19**, 1116 (2023).